

- 9 (a) An investigation was carried out to study the effect of different intensities of blue light on the percentage germination of barley seeds. Barley seeds were exposed to blue light for a period of seven days. All other variables were kept constant.

The results are shown in Table 9.1.

Table 9.1

light intensity /arbitrary units (au)	percentage germination
0 (dark)	98.0
36	76.9
48	45.0
57	14.7

The effect of blue light on the concentration of abscisic acid (ABA) was also investigated. ABA concentration was measured at intervals over seven days in barley seeds exposed to blue light at an intensity of 57 arbitrary units.

The results are shown in Table 9.2.

Table 9.2

day	concentration of ABA /arbitrary units (au)
0	100
1	90
3	350
5	351
7	381

For comparison, in the dark the concentration of ABA in barley seeds fell from 100 au at the start (day 0) to 45 au on day 1 and did **not** increase from day 1 to day 7.

ABA is thought to affect gibberellin synthesis or activity.

Using the information in Table 9.1 and Table 9.2, describe the effect of blue light on the germination of barley seeds **and** suggest an explanation for this effect.

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(b) After germination, auxin is important in the growth of barley plants.

Describe **and** explain the role of auxin in cell elongation.

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